

SAPIENZA
UNIVERSITÀ DI ROMA

# Combinatorial Optimization

## Volume 6 — Specialized algorithms

Chapters 26 of the module «Combinatorial Optimization»: specialized algorithms. Definitions, results with proofs and worked examples; exercises with complete solutions are in Exercises 6.

**Fabio Furini**

fabiofurini.github.io

Department of Computer, Control and Management Engineering — Sapienza University of Rome

Online version and downloadable material:
fabiofurini.github.io/combinatorial-optimization/
github.com/fabiofurini/combinatorial-optimization

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Chapter 26

# Knapsack: branch-and-bound on the critical item

**Prerequisites**

- From the module *Integer linear programming*, the chapter «LP-based branch-and-bound algorithms»: branching and bounding operations, the problem list, the pruning criteria, the lower bound LB and the upper bound UB. The general algorithm is not repeated here.
- From this module, the chapter «Knapsack formulations»: the models $\text{ILP}_{\text{KP}}$ and $\text{LP}_{\text{KP}}$, the efficiency of an item, the split item $s$, the residual capacity $rc$ and the bound of Dantzig.

**Objectives**

At the end of this chapter you will be able to:

- write the problem $\text{LP}^k_{\text{KP}}$ of a node, and solve it with the formulas of Dantzig, without the simplex algorithm;
- branch on the variable of the split item, unselecting or selecting that item;
- execute the branch-and-bound algorithm for the KP by hand, with the FIFO or the LIFO rule;
- explain why the algorithm is simpler than the general one, and what stays the same.

## 26.1 Why a branch-and-bound algorithm for the KP?

- **A challenge.** A knapsack of capacity 9 and six items with profits $(6, 5, 8, 9, 6, 7)$ and weights $(2, 3, 6, 7, 5, 9)$. Find the optimal value of the LP relaxation *without* the simplex algorithm. Hint: think of the efficiency $\frac{p_j}{w_j}$ of each item.
- The general algorithm solves the LP problem of every node with the dual simplex algorithm. For the KP this is not needed: the LP problem of every node is again a knapsack LP problem, solved by sorting the items by efficiency. This chapter shows what the general algorithm becomes in this case.

## 26.2 The algorithm for the KP

- Chosen a branching variable $x_j$ associated to an item $j \in \mathscr{I}$, the branching constraints of the branch-and-bound algorithm to solve the $\text{ILP}_{\text{KP}}$ model have the following form:

$$x_j \leq 0 \quad \text{and} \quad x_j \geq 1$$

  They have the effect of **unselecting** or **selecting** an item.

In a branching node $k$ of a branch-and-bound tree, we have:

1. a subset of **unselected items** $\rightsquigarrow$ denoted $\mathscr{I}^k_{\text{out}}$
2. a subset of **selected items** $\rightsquigarrow$ denoted $\mathscr{I}^k_{\text{in}}$
3. a subset of **remaining items** $\rightsquigarrow$ denoted $\mathscr{I}^k := \mathscr{I} \setminus \left\{\mathscr{I}^k_{\text{out}} \cup \mathscr{I}^k_{\text{in}}\right\}$

- A $\text{LP}_{\text{KP}}$ problem associated to a branching node $k$, called the $\text{LP}^k_{\text{KP}}$ problem, reads as follows:

$$\begin{aligned}
\max \quad & \sum_{j \in \mathscr{I}} p_j \, x_j \\
\text{subject to} \quad & \sum_{j \in \mathscr{I}} w_j \, x_j \leq c, \\
& x_j \geq 1, && \forall j \in \mathscr{I}^k_{\text{in}}, \\
& x_j \leq 0, && \forall j \in \mathscr{I}^k_{\text{out}}, \\
& x_j \leq 1, && \forall j \in \mathscr{I}, \\
& x_j \geq 0, && \forall j \in \mathscr{I}.
\end{aligned}$$

A $\text{LP}^k_{\text{KP}}$ problem of a branching node $k$ can be equivalently re-written as:

$$\underbrace{\sum_{j \in \mathscr{I}^k_{\text{in}}} p_j}_{=p(k)} + \max \quad \sum_{j \in \mathscr{I}^k} p_j \, x_j \tag{26.1a}$$

$$\text{subject to} \quad \sum_{j \in \mathscr{I}^k} w_j \, x_j \leq \underbrace{c - \sum_{j \in \mathscr{I}^k_{\text{in}}} w_j}_{=c(k)}, \tag{26.1b}$$

$$x_j \leq 1, \qquad \forall j \in \mathscr{I}^k, \tag{26.1c}$$

$$x_j \geq 0, \qquad \forall j \in \mathscr{I}^k. \tag{26.1d}$$

- A $\text{LP}^k_{\text{KP}}$ problem of a branching node $k$ is **structurally equivalent** to the $\text{LP}_{\text{KP}}$ problem but:
  1. the set of items is $\mathscr{I}^k$ instead of $\mathscr{I}$
  2. the capacity is reduced to $c(k) := c - \sum_{j \in \mathscr{I}^k_{\text{in}}} w_j$
  3. there is an objective-function constant $p(k) := \sum_{j \in \mathscr{I}^k_{\text{in}}} p_j$

### 26.2.1 Solving the $\text{LP}^k_{\text{KP}}$ problem of a node

- The problem $\text{LP}^k_{\text{KP}}$ is a knapsack LP problem on the items of $\mathscr{I}^k$, with capacity $c(k)$: it is solved by the formulas of Dantzig, without any simplex iteration. The items of $\mathscr{I}^k$ are kept in the order of non-increasing efficiency.
- If $c(k) < 0$ the problem is **infeasible**. If $\sum_{j \in \mathscr{I}^k} w_j \leq c(k)$ all the items of $\mathscr{I}^k$ fit, and the optimal solution is $\tilde{x}_j = 1$ for all $j \in \mathscr{I}^k$: it is integer.
- Otherwise the **split item** $s$ of the node is the first item of $\mathscr{I}^k$ that does not fit entirely, and the **residual capacity** is the capacity left by the items that precede it:

$$\sum_{j \in \mathscr{I}^k, j<s} w_j \leq c(k) < \sum_{j \in \mathscr{I}^k, j \leq s} w_j, \qquad rc = c(k) - \sum_{j \in \mathscr{I}^k, j<s} w_j.$$

- The optimal solution is $\tilde{x}_j = 1$ for the items of $\mathscr{I}^k$ before $s$, $\tilde{x}_s = \frac{rc}{w_s}$ and $\tilde{x}_j = 0$ for the items after $s$, and the optimal value is

$$z(\text{LP}^k_{\text{KP}}) \;=\; p(k) + \sum_{j \in \mathscr{I}^k, j<s} p_j + p_s \, \frac{rc}{w_s}.$$

In every node at most one variable is fractional, the one of the split item, and the value of the node is the Dantzig bound of a smaller knapsack: the node is solved by a scan of the items.

### 26.2.2 Branching-variable selection strategy

- The optimal solution of an $\mathrm{LP}_{\mathrm{KP}}^k$ problem given by the formulas of the previous subsection has at most one fractional variable, the one associated to the split item. **Branching can be performed on the variable associated to the split item**: this branching-variable selection strategy is called split-item strategy.
- The optimal solution of an $\mathrm{LP}_{\mathrm{KP}}$ problem is not unique when some items have equal efficiency: the other optimal solutions can have more fractional variables, for instance $x = (\frac{1}{2}, \frac{1}{2})$ for $p = (2, 2)$, $w = (2, 2)$, $c = 2$. Here, as in the general algorithm, we always use the solution given by the formulas.

### 26.2.3 What changes with respect to the general algorithm

- The algorithm is the LP-based branch-and-bound algorithm of the module *Integer linear programming*, with its problem list, its pruning criteria and its optimality gap; its pseudocode is not repeated.
- **Bounding.** The problem of a node is solved by the formulas above. All the profits are integer, so the bound of a node is $\lfloor z(\mathrm{LP}_{\mathrm{KP}}^k) \rfloor$: the node is pruned by bound if $\lfloor z(\mathrm{LP}_{\mathrm{KP}}^k) \rfloor \leq \mathrm{LB}$, by infeasibility if $c(k) < 0$, by integrality if its solution is integer (then LB is updated when the value is larger).
- **Branching.** The branching variable is the one of the split item $s$: the children are $x_s \leq 0$ (down, item unselected) and $x_s \geq 1$ (up, item selected), and the node is not solved with a simplex tableau but described by the sets $\mathscr{I}_{\mathrm{in}}^k$, $\mathscr{I}_{\mathrm{out}}^k$ and $\mathscr{I}^k$.
- **Node selection.** The node of the problem list is extracted with the FIFO or the LIFO rule, and the down-branching node is solved before the up-branching node; the nodes are numbered in the order of resolution. The lower bound written in a node is the value at the moment of its resolution.

## 26.3 A worked example

**Example 26.1: The $\mathrm{ILP}_{\mathrm{KP}}$ model**

We consider a knapsack of capacity $c = 9$ and a set $\mathscr{I} = \{1, 2, 3, 4, 5, 6\}$ of 6 items ($n = 6$), with the following profits and weights:

| | $j=1$ | $j=2$ | $j=3$ | $j=4$ | $j=5$ | $j=6$ |
|---|---|---|---|---|---|---|
| $p_j$ | 6 | 5 | 8 | 9 | 6 | 7 |
| $w_j$ | 2 | 3 | 6 | 7 | 5 | 9 |

We refer to this KP instance as Example 1. The $\mathrm{ILP}_{\mathrm{KP}}$ model for this KP instance is:

$$
\begin{array}{rrlrlrlrlrlrlll}
\max & 6 & x_1 & +5 & x_2 & +8 & x_3 & +9 & x_4 & +6 & x_5 & +7 & x_6 & & \\
\text{subject to} & 2 & x_1 & +3 & x_2 & +6 & x_3 & +7 & x_4 & +5 & x_5 & +9 & x_6 & \leq & 9 \\
& & x_1, & & x_2, & & x_3, & & x_4, & & x_5, & & x_6 & \in & \{0,1\}
\end{array}
$$

An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{ILP_{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 1, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$

The optimal value $z(\mathrm{ILP_{KP}})$ is 15. The associated optimal solution of the KP is $\tilde{S} = \{1, 4\}$.

The items are sorted by non-increasing efficiency, since:

$$\frac{6}{2} \geq \frac{5}{3} \geq \frac{8}{6} \geq \frac{9}{7} \geq \frac{6}{5} \geq \frac{7}{9}$$

**Example 26.2: Branch-and-bound algorithm for the $\mathrm{ILP_{KP}}$ model**

A branch-and-bound algorithm for $\mathrm{ILP_{KP}}$ model and the KP instance of Example 1 using:

1. the split-item strategy as branching-variable selection strategy;
2. the FIFO rule as branching-node selection strategy;
3. solve first the down-branching node and then the up-branching node;
4. set the incumbent solution $\bar{\boldsymbol{x}}(\mathrm{ILP_{KP}})$ to the following heuristic solution:

   $$\bar{x}_1 = 1, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1, \quad \bar{x}_5 = 0, \quad \bar{x}_6 = 0$$

   and set the lower bound LB accordingly.

- **Initialization**

  For the $\mathrm{LP^1_{KP}}$ problem, the split item is $s = 3$ and the residual capacity is $rc = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP^1_{KP}}\,)$ is:

  $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{4}{6}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$

  the optimal value is:

  $$z(\,\mathrm{LP^1_{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 8 \cdot \frac{4}{6} = \frac{49}{3}$$

  The optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP^1_{KP}}\,)$ is **fractional** and the $\mathrm{LP^1_{KP}}$ problem is **feasible**, so the $\mathrm{LP^1_{KP}}$ problem is inserted in the problem list $L$ and the lower bound LB is set to 15 ($\mathrm{LB}(\bar{\boldsymbol{x}}) = 15$).

  Due to the current lower bound LB and upper bound UB, we have:

  $$15 \quad \leq \quad z(\mathrm{ILP_{KP}}) \quad \leq \quad \underbrace{\left\lfloor \frac{49}{3} \right\rfloor}_{=16}$$

- **Iteration 1**

  The $\mathrm{LP}^1_{\mathrm{KP}}$ problem is extracted and deleted from the problem list $L$.

  The branching variable is $x_3$. The $\mathrm{LP}^2_{\mathrm{KP}}$ and $\mathrm{LP}^3_{\mathrm{KP}}$ problems are created by unselecting $(x_3 \leq 0)$ and selecting $(x_3 \geq 1)$ item 3.

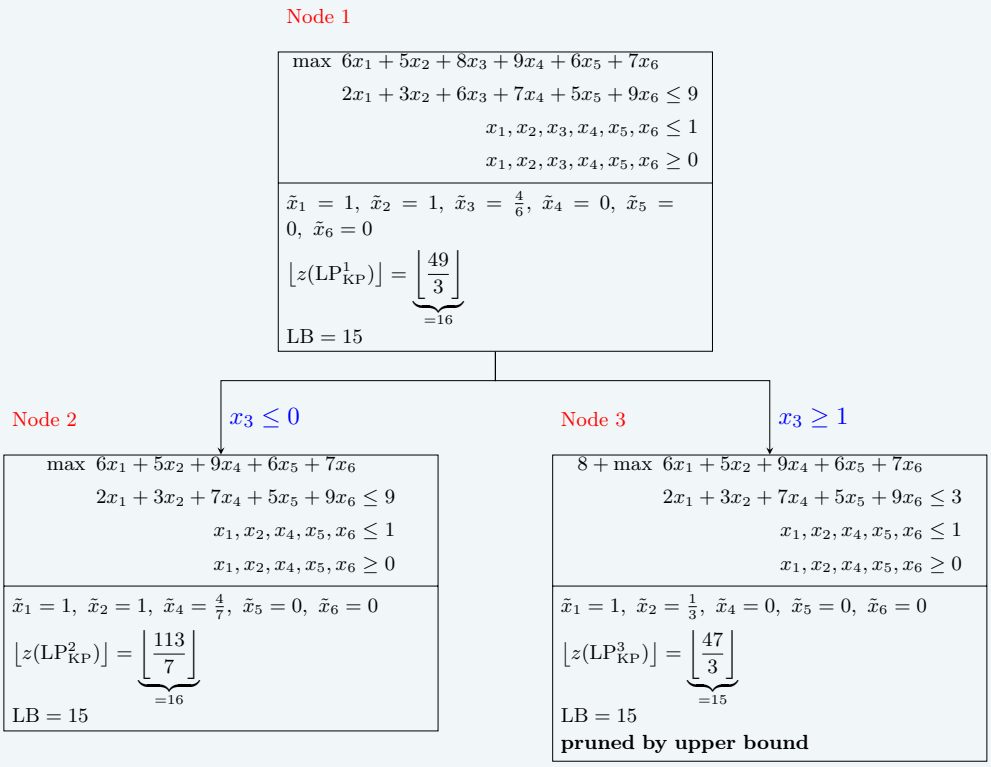
  1. Solution of the $\mathrm{LP}^2_{\mathrm{KP}}$ problem

     We have
     $$\mathscr{I}^2 = \{1,2,4,5,6\}, \quad \mathscr{I}^2_{\mathrm{out}} = \{3\}, \quad \mathscr{I}^2_{\mathrm{in}} = \emptyset$$
     The capacity is $c(2) = 9$ and the objective-function constant is $p(2) = 0$. The split item is $s = 4$ and the residual capacity is $rc = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^2_{\mathrm{KP}}\,)$ is:
     $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_4 = \frac{4}{7}, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$
     and
     $$z(\,\mathrm{LP}^2_{\mathrm{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 9 \cdot \frac{4}{7} = \frac{113}{7}$$
     The $\mathrm{LP}^2_{\mathrm{KP}}$ problem is **feasible** and we have:
     $$\lfloor z(\mathrm{LP}^2_{\mathrm{KP}}) \rfloor > \mathrm{LB} \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{113}{7} \right\rfloor}_{=16} > 15$$
     The optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^2_{\mathrm{KP}}\,)$ is **fractional**, accordingly, the $\mathrm{LP}^2_{\mathrm{KP}}$ problem is **inserted in the problem list** $L$.

  2. Solution of the $\mathrm{LP}^3_{\mathrm{KP}}$ problem

     We have
     $$\mathscr{I}^3 = \{1,2,4,5,6\}, \quad \mathscr{I}^3_{\mathrm{out}} = \emptyset, \quad \mathscr{I}^3_{\mathrm{in}} = \{3\}$$
     The capacity is $c(3) = 3$ and the objective-function constant is $p(3) = 8$. The split item is $s = 2$ and the residual capacity is $rc = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^3_{\mathrm{KP}}\,)$ is:
     $$\tilde{x}_1 = 1, \ \tilde{x}_2 = \frac{1}{3}, \ \tilde{x}_4 = 0, \ \tilde{x}_5 = 0, \ \tilde{x}_6 = 0 \quad \text{and} \quad z(\,\mathrm{LP}^3_{\mathrm{KP}}\,) = 8{+}6{\cdot}1{+}5{\cdot}\frac{1}{3} = \frac{47}{3}$$
     The $\mathrm{LP}^3_{\mathrm{KP}}$ problem is **feasible** and we have:
     $$\lfloor z(\mathrm{LP}^3_{\mathrm{KP}}) \rfloor \leq \mathrm{LB} \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{47}{3} \right\rfloor}_{=15} \leq 15$$
     accordingly, the branching node is pruned by upper bound.

At the end of the first iteration, Node 2 is inserted in the problem list $L$, while Node 3 is pruned by upper bound. The corresponding branch-and-bound tree is:




Due to the current lower bound LB and upper bound UB, we have:

$$15 \ \le \ z(\mathrm{ILP}_{\mathrm{KP}}) \ \le \ \underbrace{\left\lfloor \frac{113}{7}\right\rfloor}_{=16}$$

- **Iteration 2**

The $\mathrm{LP}^2_{\mathrm{KP}}$ problem is extracted and deleted from the problem list $L$.

The branching variable is $x_4$. The $\mathrm{LP}^4_{\mathrm{KP}}$ and $\mathrm{LP}^5_{\mathrm{KP}}$ problems are created by unselecting $(x_4 \le 0)$ and selecting $(x_4 \ge 1)$ item 4.

1. Solution of the $\mathrm{LP}^4_{\mathrm{KP}}$ problem

   We have

$$\mathscr{I}^4 = \{1,2,5,6\}, \quad \mathscr{I}^4_{\mathrm{out}} = \{3,4\}, \quad \mathscr{I}^4_{\mathrm{in}} = \emptyset$$

The capacity is $c(4) = 9$ and the objective-function constant is $p(4) = 0$. The split item is $s = 5$ and the residual capacity is $rc = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^4_{\mathrm{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_5 = \frac{4}{5}, \quad \tilde{x}_6 = 0 \quad \text{and} \quad z(\,\mathrm{LP}^4_{\mathrm{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 6 \cdot \frac{4}{5} = \frac{79}{5}$$

The $\mathrm{LP}^4_{\mathrm{KP}}$ problem is **feasible** and we have:

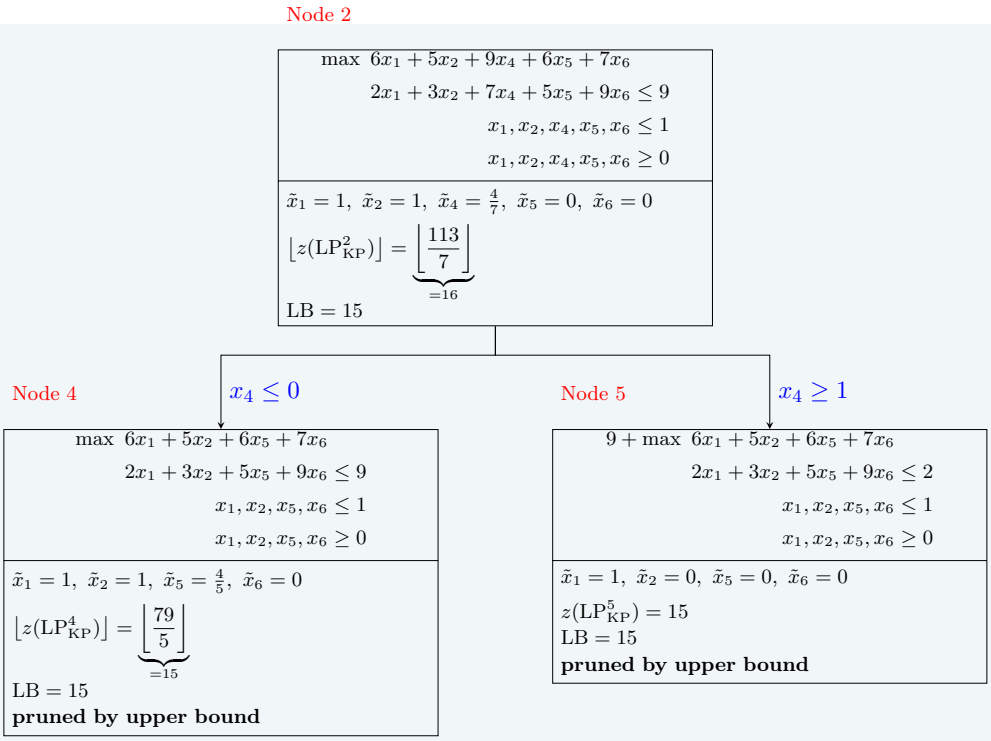
$$\lfloor z(\mathrm{LP}^4_{\mathrm{KP}}) \rfloor \leq \mathrm{LB} \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{79}{5} \right\rfloor}_{=15} \leq 15$$

accordingly, the branching node is pruned by upper bound.

2. Solution of the $\mathrm{LP}^5_{\mathrm{KP}}$ problem

   We have
   $$\mathscr{I}^5 = \{1, 2, 5, 6\}, \quad \mathscr{I}^5_{\mathrm{out}} = \{3\}, \quad \mathscr{I}^5_{\mathrm{in}} = \{4\}$$
   The capacity is $c(5) = 2$ and the objective-function constant is $p(5) = 9$. The split item is $s = 2$ and the residual capacity is $rc = 0$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^5_{\mathrm{KP}}\,)$ is:
   $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0 \quad \text{and} \quad z(\,\mathrm{LP}^5_{\mathrm{KP}}\,) = 9 + 6 \cdot 1 = 15$$
   The $\mathrm{LP}^5_{\mathrm{KP}}$ problem is **feasible** and we have:
   $$\lfloor z(\mathrm{LP}^5_{\mathrm{KP}}) \rfloor \leq \mathrm{LB} \qquad \text{since} \qquad 15 \leq 15$$
   accordingly, the branching node is pruned by upper bound. The optimal solution of the $\mathrm{LP}^5_{\mathrm{KP}}$ problem is also integer, but its value does not improve the incumbent solution: the node is pruned by bound, as the other nodes of this example.

At the end of the second iteration, Node 4 and Node 5 are both pruned by upper bound. The corresponding branch-and-bound tree is:

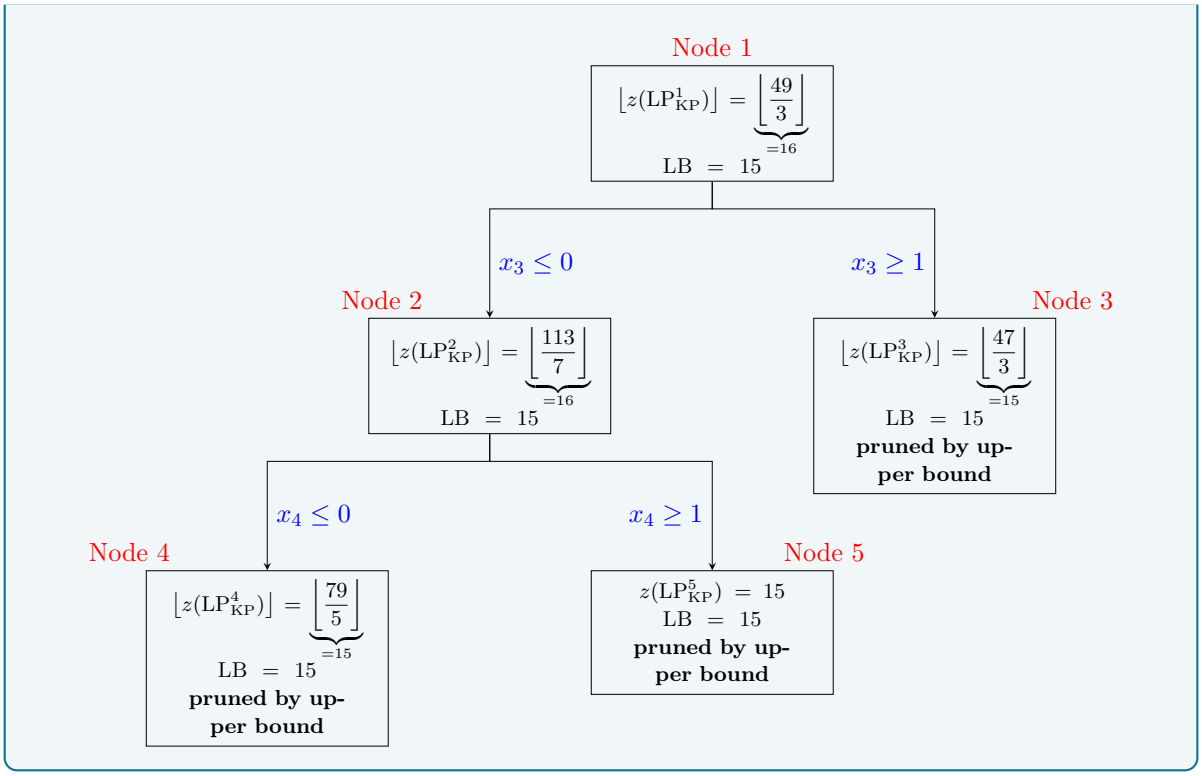



- **Termination**

  Since the problem list $L$ is empty, the $\mathrm{ILP}_{\mathrm{KP}}$ problem is solved to optimality.

  The optimal value is $z(\mathrm{ILP}_{\mathrm{KP}}) = 15$ (the value of the lower bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ \mathrm{ILP}_{\mathrm{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1, \tilde{x}_5 = 0, \tilde{x}_6 = 0$ (the incumbent solution $\bar{\boldsymbol{x}}(ILP_{\mathrm{KP}})$).

This is the branch-and-bound tree of the previous B&B Algorithm, showing for each branching node the associated $\mathrm{LP}_{\mathrm{KP}}$ problem, its optimal solution and value, the lower bound LB, and the pruning criteria.




## 26.4 Typical mistakes

**Warning**

- **Forgetting the upper bounds $x_j \leq 1$.** Without them the LP problem of a node would put all the capacity on the item of best efficiency: the node boxes must contain $x_j \leq 1$.
- **Computing the split item on the original instance.** In a node, the split item and the residual capacity are computed on the items of $\mathscr{I}^k$ with the capacity $c(k)$, after fixing the items of $\mathscr{I}^k_{\mathrm{in}}$ and $\mathscr{I}^k_{\mathrm{out}}$.
- **Forgetting the constant $p(k)$.** The value of the node is $p(k)$ plus the value of the items of $\mathscr{I}^k$: the profit of the selected items counts.
- **Using the old lower bound.** The lower bound is updated as soon as a node has an integer solution of larger value; the nodes of the list that become dominated are discarded.
- **Numbering the skipped children.** A child that is not solved does not receive a node index, exactly as in the general algorithm.

## 26.5 In summary

**Summary**

- The branch-and-bound algorithm for the KP is the LP-based algorithm with two simplifications: the problem of every node is solved by the formulas of Dantzig, and the branching variable is the one of the split item.
- A node is described by the sets $\mathscr{I}^k_{\text{in}}$, $\mathscr{I}^k_{\text{out}}$ and $\mathscr{I}^k$, by the capacity $c(k)$ and by the constant $p(k)$.
- The bound of a node is $\lfloor z(\text{LP}^k_{\text{KP}}) \rfloor$; when the problem list is empty the lower and the upper bound coincide and the incumbent solution is optimal.

**Dashboard of the KP instance of Example 1** ($c = 9$, $p = (6, 5, 8, 9, 6, 7)$, $w = (2, 3, 6, 7, 5, 9)$). The bounds on $z(\text{ILP}_{\text{KP}})$ found so far:

| Source | Lower bound | Upper bound | From |
|---|---|---|---|
| Dantzig bound | – | 16 | $\lfloor \frac{49}{3} \rfloor$ |
| This chapter, initialization | 15 | 16 | heuristic solution $\{1, 4\}$ |
| This chapter, iteration 1 | 15 | 16 | node 2 in the list, $\lfloor \frac{113}{7} \rfloor$ |
| This chapter, termination | 15 | 15 | empty list, 5 nodes |